MODELLING OF MECHANICAL STRESS DEGRADATION AND SEI LAYER GROWTH IN LI ION BATTERIES

KIRUTIGA S, VENKATASAILANATHAN RAMADESIGAN
Department of Energy Science and Engineering, IIT Bombay

INTRODUCTION

- Li batteries – energy storage systems.
- Predict and monitor degradation – Enhanced battery performance.
- Physics based model – superior over empirical models – Predict SEI Layer growth
- Quantify diffusion induced stress
- Type of battery – Graphite /LMO cell

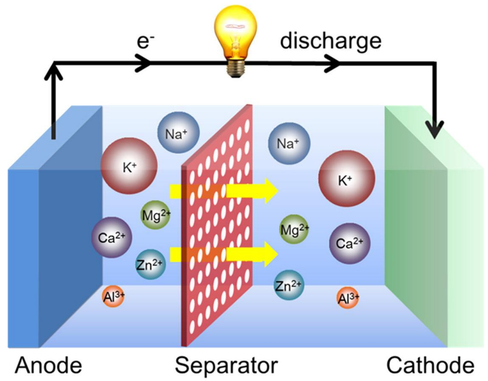


Fig 1: Schematic representation of a battery [1]

OBJECTIVE

- To investigate and quantify stress due to mechanical degradation

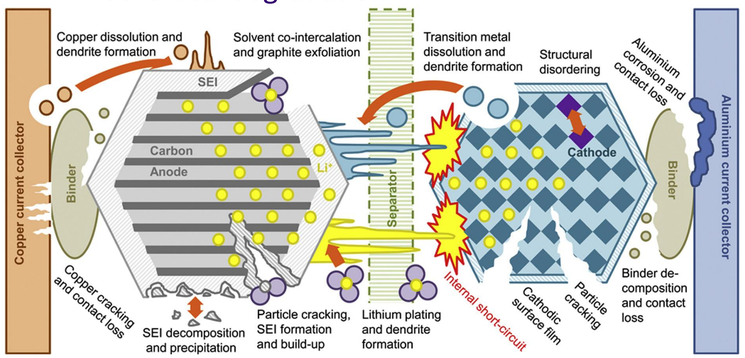
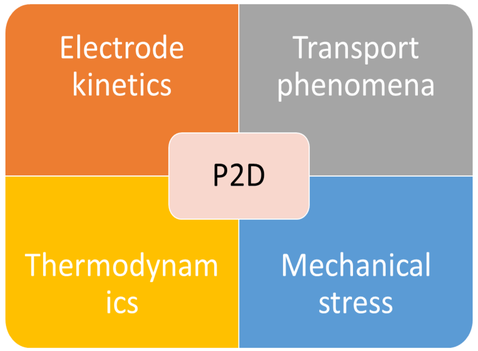


Fig 2: Illustration of degradation mechanisms [2]

MODEL DESCRIPTION

- Equations for Conservation of **charge** and **species**
- P2D model – Coupled Partial Differential Equations – Set of 10 PDEs [3]
- Accounts for transport and kinetics in porous electrodes

GOVERNING EQUATIONS



- Butler Volmer kinetics used to model unwanted side reaction – SEI Layer growth

$$j_{SEI} = k_{SEI} \exp\left(\frac{\alpha}{FRT}\left(\varphi_1 - \varphi_2 - U_{SEI} - \frac{\delta}{k_{SEI}}(j_n + j_{SEI})\right)\right)$$

- Mechanical degradation – Diffusion induced stress

$$\frac{\partial C}{\partial t} = (1+\theta C)\frac{\partial^2 C}{\partial r^2} + \left(\frac{2}{r} + \theta\frac{\partial C}{\partial r} + \frac{2\theta C}{r}\right)\frac{\partial C}{\partial r}$$

- Stress generated due to particle intercalation is considered.

( )

( )

RESULTS

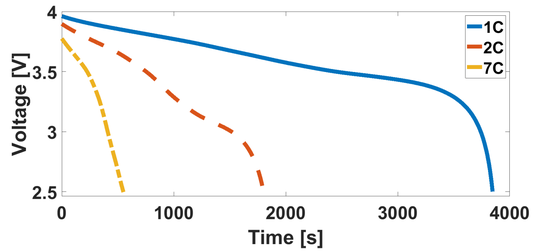


Fig 3: Voltage vs Time under constant input power density

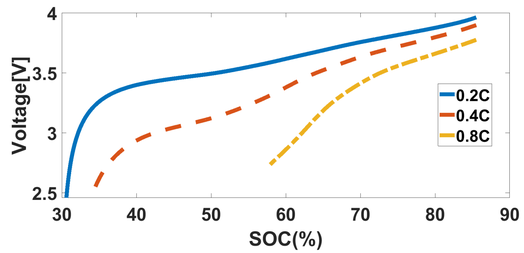


Fig 5: Voltage- SOC characteristics of the battery

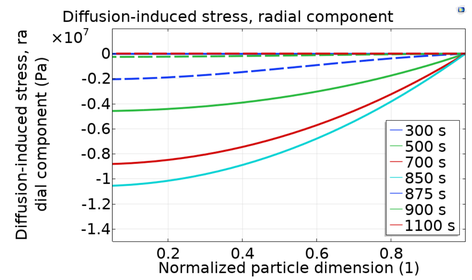


Fig 7: Diffusion induced stress, radial component

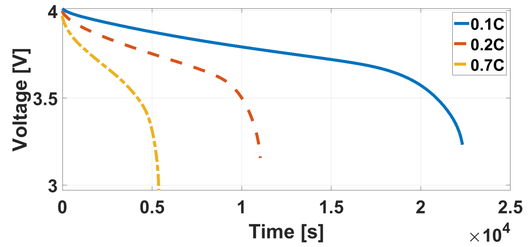


Fig 4: Voltage vs Time under constant input current density

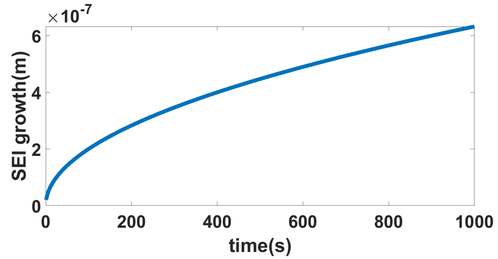


Fig 6: SEI layer growth with time

SUMMARY

- The diffusion induced stress generated is quantified.
- The SEI layer growth follows the parabolic law proposed by Peled et al.[4]
- The negative stress value indicates the compressive nature of stress
- Voltage characteristics show different rate of discharge under different operating conditions.

FUTURE WORK

- Integration of chemical and mechanical degradation mechanisms – battery pack.
- Addressing capacity fade in EV batteries – Increase economic value of Li ion batteries.

ACKNOWLEDGEMENT

IRCC Research Internship Awards 2018-2019, IITB

REFERENCES

[1] Y. Wang *et al.*, "Emerging non-lithium ion batteries," *Energy Storage Materials*, vol. 4, pp. 103–129, Jul. 2016.
[2] C. R. Birkl, M. R. Roberts, E. McTurk, P. G. Bruce, and D. A. Howey, "Degradation diagnostics for lithium ion cells," *Journal of Power Sources*, vol. 341, pp. 373–386, Feb. 2017.
[3] M. Torchio, L. Magni, B. G. R, R. D. Braatz, and D. M. Raimondo, "LIONSIMBA: A Matlab Framework Based on a Finite Volume Model Suitable for Li-Ion Battery Design, Simulation, and Control," *J. Electrochem. Soc.*, vol. 163, no. 7, pp. A1192–A1205.
[4] E. Peled and S. Menkin, "Review—SEI: Past, Present and Future," *J. Electrochem. Soc.*, vol. 164, no. 7, pp. A1703–A1719, Jan. 2017